

TECHNICAL PROPOSAL



Project DRISHTI

AI-Based Granite Block Measurement and Automated Tax Assessment

Computer Vision for Transparent
Mineral Revenue Governance

Prepared for: **RTGS, Government of Andhra Pradesh**

Challenge: Smart Granite Block Measuring & Tax Assessment

Hackathon: RTGS AI Hackathon 2026

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SECTION 1: Executive Summary

Project DRISHTI is an AI-powered platform that automatically measures granite block dimensions using computer vision, estimates volume, classifies granite type, and calculates applicable government seigniorage fees—replacing subjective manual measurements with standardized, auditable digital assessment.

Granite is one of Andhra Pradesh’s most valuable mineral resources. Current field inspection processes rely on manual measurements by DMGO officers—subjective, inconsistent, and prone to revenue leakage. DRISHTI transforms this with GPS-enabled mobile image capture, AI-powered dimension estimation, and automated fee calculation integrated with OMEPS 2.0.

Key capabilities include:

- AI-powered granite block detection and dimension estimation from mobile images
- Automated volume calculation with confidence scoring
- Granite classification (Above/Below Gangsaw) and seigniorage fee computation
- GPS-tagged, timestamp-verified mobile capture ensuring authenticity
- Integration-ready REST APIs for OMEPS 2.0
- Complete digital audit trail from quarry measurement to dispatch
- Cross-validation with weighbridge records for anomaly detection

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Focus Area	AI, Computer Vision & Digital Governance

SECTION 2: Problem Statement

Andhra Pradesh's granite mining sector generates significant government revenue through seigniorage fees. However, the current assessment process has critical weaknesses:

- **Subjective manual measurements:** Dimension estimation varies between officers, leading to inconsistent fee assessment across quarry locations.
- **Revenue leakage:** Inaccurate volume calculations directly impact seigniorage collection. Under-estimation results in government revenue loss.
- **No standardization:** No validated mechanism exists for cross-checking quarry measurements against weighbridge data or dispatch records.
- **High inspection overhead:** Field inspections consume significant administrative resources. Each quarry requires physical visits for block-by-block measurement.
- **No digital traceability:** Manual workflows lack linkage between measured blocks, weigh-bridge records, and dispatch details—making post-inspection audit difficult.
- **Slow turnaround:** Manual measurement, documentation, and approval workflows increase time from extraction to dispatch clearance.

SECTION 3: Solution Overview

DRISHTI introduces four integrated components:

3.1 AI Granite Measurement Engine

Computer vision pipeline that detects granite blocks in mobile-captured images, estimates dimensions (Length $\times$ Width $\times$ Height), calculates volume, and generates confidence scores:

- YOLOv11 for granite block detection and segmentation
- Depth Anything V2 for monocular depth estimation
- Perspective correction and geometric analysis for dimension computation
- Reference calibration (known-size marker) for metric accuracy

3.2 Mobile Field Application

Flutter-based mobile app for quarry operators and DMGO officers:

- GPS-tagged image capture with timestamp validation
- Camera calibration guidance for standardized photography
- Offline capture with secure synchronization
- Block ID entry and quarry metadata recording

3.3 Automated Seigniorage Engine

Rules engine that maps AI-computed dimensions to applicable government fee slabs:

- Volume calculation from estimated dimensions
- Above/Below Gangsaw classification based on block size
- Granite category mapping per government schedule
- Seigniorage fee computation with detailed assessment report

3.4 Digital Governance Platform

Administrative layer providing audit trail and analytics:

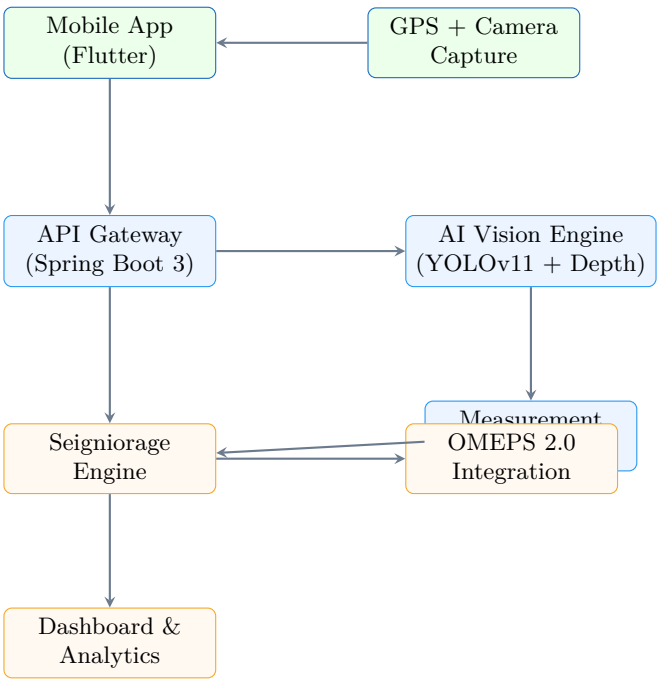
- Block-level digital lifecycle record (measurement $\rightarrow$ classification $\rightarrow$ fee $\rightarrow$ dispatch)
- Cross-validation with weighbridge and transport records
- Dashboard with quarry-wise, district-wise analytics
- OMEPS 2.0 integration via secure REST APIs

SECTION 4: Technical Architecture

4.1 Technology Stack

Layer	Technology
Mobile App	Flutter (Android + iOS)
AI Models	YOLOv11 (block detection), Depth Anything V2 (depth estimation), OpenCV (geometric analysis)
Backend	Java 21, Spring Boot 3
Database	PostgreSQL
Cloud Storage	Amazon S3
API Security	JWT, OAuth2
Deployment	Docker, AWS ECS
Monitoring	Prometheus, Grafana

4.2 Architecture Diagram



SECTION 5: AI Measurement Workflow

The computer vision pipeline processes field photographs through six stages:

1. **Image Quality Validation:** Checks blur, exposure, resolution before processing. Rejects unusable images with guidance for retake.
2. **Granite Block Detection:** YOLOv11 identifies and segments individual granite blocks from background (ground, equipment, other blocks).
3. **Depth Estimation:** Depth Anything V2 generates monocular depth map. Combined with reference calibration marker for metric scaling.
4. **Dimension Estimation:** Geometric analysis extracts Length, Width, and Height from depth map and perspective geometry. Multiple viewpoints improve accuracy.
5. **Volume Calculation:** Computed as $L \times W \times H$ with confidence interval. Handles irregular shapes via convex hull approximation.
6. **Classification & Fee:** Volume mapped to Above/Below Gangsaw category. Applicable seigniorage slab identified per government schedule. Fee calculated automatically.

SECTION 6: User Journeys

Quarry Operator: Captures standardized photos of freshly extracted granite blocks using the DRISHTI mobile app. GPS coordinates, Block ID, and quarry details auto-captured. Receives instant AI-estimated dimensions and indicative seigniorage fee. Submits for officer verification.

DMGO Officer: Reviews AI-computed measurements on dashboard. Compares with visual evidence (photos). For high-confidence predictions (>90%), approves directly. For lower confidence, visits site for verification. Reduces physical inspection frequency by 70%.

Mining Department: Views district-wise seigniorage analytics. Identifies quarries with anomalous measurement patterns (AI vs. weighbridge discrepancy). Triggers audit for revenue recovery. Tracks monthly collection trends and forecasts.

SECTION 7: Success Metrics

- Granite dimension estimation accuracy >**95%** (within 5cm of actual measurement).
- Volume estimation accuracy >**90%** (validated against manual measurement baseline).
- AI confidence score >**85%** for 80%+ of blocks photographed under standard conditions.
- Physical inspection reduction of **70%** for routine measurements.
- Processing time <**30 seconds** from image upload to dimension result.
- OMEPS 2.0 integration operational with **100%** data synchronization.

- Revenue leakage reduction—measurable improvement in seigniorage collection per quarry.

SECTION 8: Risks and Mitigations

- **Variable quarry conditions:** Dust, uneven lighting, and wet surfaces affect image quality. Mitigate with real-time quality validation, LED flash guidance, and model training on diverse field conditions.
- **Irregular block shapes:** Not all blocks are perfect cuboids. Use convex hull volume estimation and flag irregular blocks for manual measurement with lower confidence score.
- **GPS spoofing:** Fake locations could enable fraud. Implement multi-factor validation: GPS + cell tower triangulation + photo metadata cross-check + timestamp sequencing.
- **Network connectivity at quarries:** Remote quarries may lack internet. Offline capture with secure local storage; auto-sync when connectivity returns. Tamper-evident local storage prevents data modification.
- **Calibration accuracy:** Reference marker essential for metric accuracy. Standard government-issued calibration boards placed alongside blocks. Model detects missing/damaged markers.

SECTION 9: Why PeopleWave

PeopleWave combines computer vision expertise with cloud-native architecture and AP governance domain knowledge. We have delivered AI solutions for RTIH and understand the integration requirements of government systems like OMEPS 2.0.

For Project DRISHTI, our team brings experience in mobile AI deployment, depth estimation models, and GPS-validated field data capture—all critical for a solution that must work reliably at quarry sites across Andhra Pradesh's diverse terrain and conditions.

DRISHTI is designed not as a hackathon demo but as a deployment-ready solution capable of supporting statewide granite measurement and transparent mineral revenue governance.

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